
Microeconomics
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L3

Elasticity

Size of Changes in Market Equilibrium

- What determines the size of changes in market equilibrium?
 - Size of change in demand (or supply)
 - The larger the shift in demand (or supply), the larger the effect on the equilibrium P and Q .
 - Steepness of the curve that does not shift
 - Steepness reflects responsiveness of Q demanded to prices

Elasticity of Y to X

- A measure of the responsiveness of Y to small changes in X
- General elasticity formula:

$$E_X^Y = \frac{\% \text{change in } Y}{\% \text{change in } X}$$

Where:

- % change in Y: $100(Y_n - Y_o)/Y_o$
- % change in X: $100(X_n - X_o)/X_o$

Interpreting an Elasticity

➤ Suppose $E=2$

➤ Then Y increases 2% for each 1% increase in X

➤ Suppose $E=-2$

➤ Then Y decreased 2% for each 1% increase in X

Elasticity of Demand

- We are interested in measuring the percentage change in the individual demand for a good for each 1% change in:
 1. The price of the good
 2. The price of other goods
 3. The consumer's income

Price elasticity of demand: E_P^d

- Measures how responsive the quantity demanded is to small changes in prices
- $E_P^d = \frac{\%change\ in\ Q}{\%change\ in\ P}$
- Expect E^d to be negative:
 - When P increases, amount demanded typically decreases
 - When P decreases, amount demanded typically increases

Categories of Elasticity of Demand

➤ ELASTIC: $|E^d| > 1$

- Meaning: 1% rise in price reduces consumption by more than 1%.

➤ INELASTIC: $|E^d| < 1$

- Meaning: a 1% rise in price reduces consumption by less than 1%

➤ UNIT ELASTIC: $|E^d| = 1$

- Meaning: a 1% rise in price reduces consumption by 1%.

Categories of Elasticity of Demand

Extreme cases:

- **PERFECTLY ELASTIC:** $E^d = -\infty$ (horizontal D curve)
 - Meaning: a rise in the price of the good induces consumers to stop purchasing it.
- **PERFECTLY INELASTIC:** $E^d = 0$ (vertical D curve)
 - Meaning: a rise in the price of the good won't affect consumption.

Price Elasticity of Demand

- Goods tend to have more price elastic demand when:
 - They have close substitutes → if price changes, consumers can easily switch to an alternative product
 - Buyers of the product consider it a luxury
 - Buyers of the product have low income and thus are sensitive to changes in their expenditures
- *In general*, elasticity of demand varies at different points along a demand curve
 - At very high prices a 2% increase will have different effect on demand than at very low prices
 - we measure elasticity separately at each point of the demand curve (i.e. at each initial price)

Price elasticity for linear demand curves

$$E^d = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in price}} = \frac{100 \times (\Delta Q/Q)}{100 \times (\Delta P/P)} = \frac{(\Delta Q/Q)}{(\Delta P/P)}$$

$$E^d = \frac{(\Delta Q/Q)}{(\Delta P/P)} = \left(\frac{\Delta Q}{Q} \right) \left(\frac{P}{\Delta P} \right) = \left(\frac{\Delta Q}{\Delta P} \right) \left(\frac{P}{Q} \right)$$

DERIVATIVE OF DEMAND FUNCTION

INITIAL P and Q

Price elasticity for linear demand curves

For a general linear demand function:

$$Q^d = a - bP$$

$$E^d = -b(P_0/Q_0)$$

Where

- $(-b)$ is the derivative of the linear demand function
- P_0 and Q_0 are the initial price and quantities

Example

- What is the elasticity of the demand function

$$Q_{corn}^d = 15 - 2P_{corn}$$

at an initial price of 3.75?

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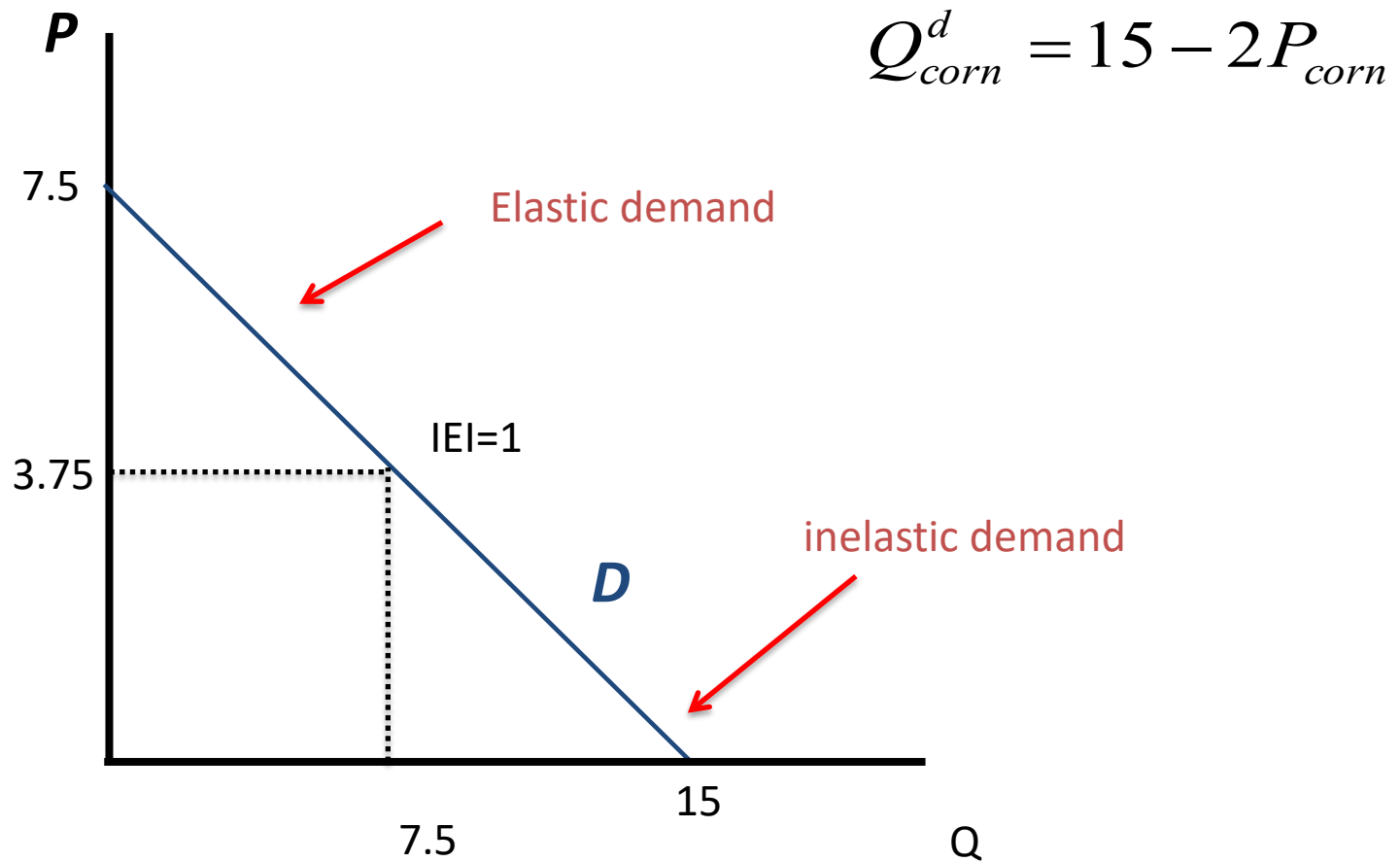
- $E^d = -b(P_0/Q_0)$:
- $Q_0 = 15 - 7.5 = 7.5$
- derivative = -2
- $E^d = -2(3.75/7.5) = -1$

Elasticities for Linear Demand Curves

➤ Notice that:

- Slope is constant along linear demand curve but (P_0/Q_0) varies, so elasticity varies along the demand curve
- Demand is more elastic at higher prices since P_0 is larger and Q_0 is smaller
- Demand is less elastic at lower prices since P_0 is smaller and Q_0 is larger
- For this reason, we measure elasticity separately at each point of the demand curve (i.e. at each initial price).

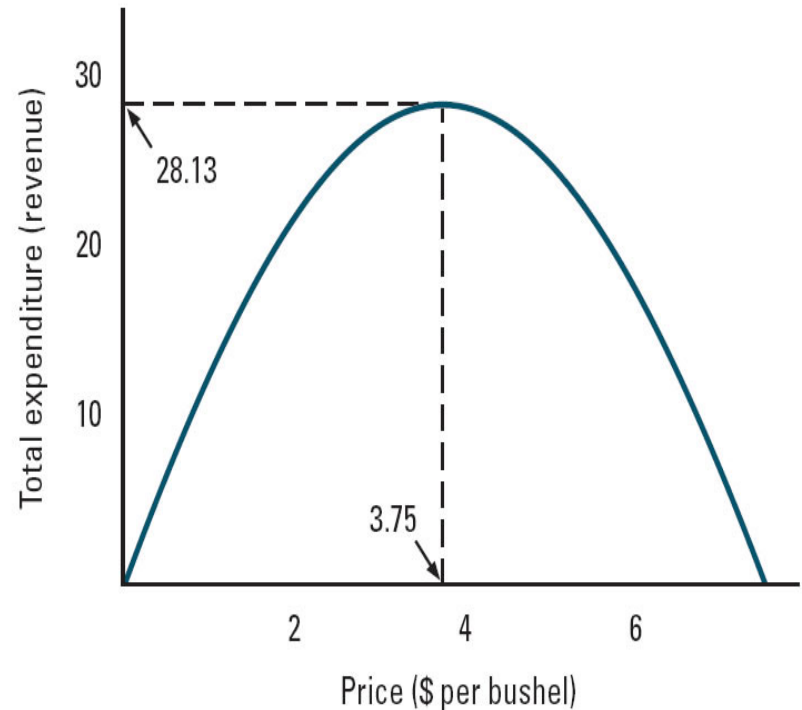
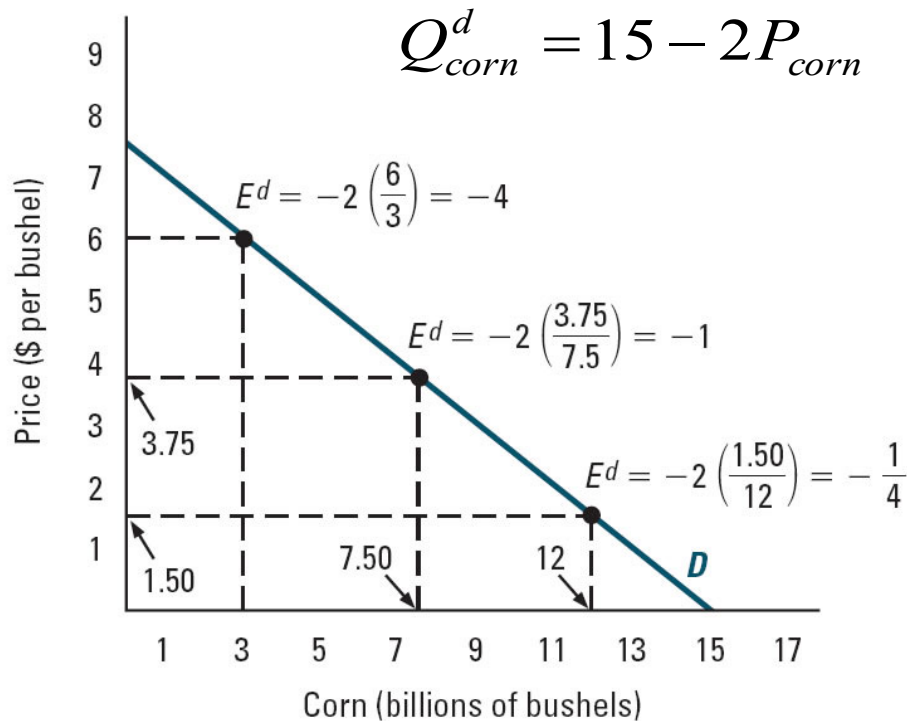
Elasticities Along a Linear Demand Curve



Total Expenditure and Elasticity of Demand

- **Total expenditure (TE)** = $P \times Q$, the product of the price and the total amount demanded
 - %change in **TE** = %change in **Q** + %change in **P**
- Elasticity of demand tells us how total expenditure changes when price decreases
- **TE** will increase with a small decrease in price when demand is elastic and decrease when demand is inelastic
- If $E^d = -1$, then a 1% increase in **P** will decrease **Q** by exactly 1%, leaving **TE** unchanged.
- **TE** is largest at a price for which elasticity equals -1

Price, Elasticity, and Total Expenditure



- TE decreases with a P cut where demand is inelastic; for prices below \$3.75
- TE increases where demand is elastic
- TE is largest where $E^d = -1$; when price = \$3.75

Income Elasticity of Demand

- Measures responsiveness of a product's demand to changes in consumers' income

$$E_M^d = \frac{\% \Delta \text{ amount demanded}}{\% \Delta \text{ income}} = \frac{(\Delta Q / Q)}{(\Delta M / M)} = (\Delta Q / \Delta M) * M_0 / Q_0$$

- E_M^d may be positive or negative depending on the good considered and on consumers' preferences over the good.

Income Elasticity of Demand

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- **Normal goods:** income elasticity > 0
- **Inferior goods:** income elasticity < 0

Cross-price Elasticity of Demand

- Measures responsiveness of the quantity demanded to changes in another product's price

$$E_{P_o}^d = \frac{\% \Delta \text{ amount demanded}}{\% \Delta \text{ other product's price}} = \frac{(\Delta Q / Q)}{(\Delta P^o / P^o)} = (\Delta Q / \Delta P^o) P_o^o / Q_o$$

- $E_{P_o}^d$ may be positive, negative or equal to zero depending on the relationship between the two goods.

Cross-price Elasticity of Demand

- Measures responsiveness of the quantity demanded to changes in another product's price

$$E_{P^o}^d = \frac{\% \Delta \text{ amount demanded}}{\% \Delta \text{ other product's price}} = \frac{(\Delta Q / Q)}{(\Delta P^o / P^o)} = (\Delta Q / \Delta P^o) P^o / Q_0$$

- **Substitute:** cross-price elasticity > 0
- **Complements:** cross-price elasticity < 0

Elasticity of Supply

- E^S measures the change in Q^S to a change in P
- $E^S = \% \text{change in } P / \% \text{change in } Q$
- We expect it to be positive
- We measure it at every point of the supply curve (that is for every initial price)
- For linear supply curves: $E^S = bP_0/Q^S_0$